Base-specificity in Lexical Conservatism

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1. Introduction

This paper presents an experimental test of a common assumption in the phonological literature on Lexical Conservatism, namely that the effects of the phonologically-advantageous stem allomorph that licenses a markedness-reducing phonological alternation in a derived form yield outcomes where the derived form specifically resembles the licensing allomorph. We report the results of a *wug*-test with adult native speakers of English that uses trisyllabic stimuli, and find that the base-specificity assumption is supported: derived forms licensed by stem allomorphs with penult stress exhibit greater propensity for penult stress than those with no licensing forms and those with final-stressed forms, and derived forms licensed by final-stress stem allomorphs exhibit more final stress than those with no licensing forms and with penult-stressed forms. We fit formal grammatical models to the experimental data implementing three different models of Lexical Conservatism that from the literature, and use model comparison techniques to quantify how much support each theory receives from the data. We find that of the three theories investigated, only the Voting Bases model proposed in Breiss (2024) is flexible enough to capture the salient characteristics of the experimental data.

1.1. Background on Lexical Conservatism

Lexical Conservatism is the name given to an empirical phenomenon first described by Steriade (1997), where the phonological form of stem allomorphs in a paradigm influences the way that paradigm accommodates novel members. For example, despite the phonological similarity of the words *cómpensate* and *ínundate*, they behave differently when affixed with *-able*, yielding stress-shifted *compénsable* but non-alternating *ínundable*. Steriade suggests that this difference is due not to properties of *cómpensate* and *ínundate* themselves — which we refer to in this paper as Local Bases — but rather due to the presence of a morphologically-related form *compénsatory*, which has a phonological characteristic — stem-final stress — that allows the otherwise unlicensed stress shift to take place in *compénsable*. We use the term Derivative to refer to the novel derived form being added to the paradigm (here *compénsable* and *ínundable*), and we term the morphologically-related form the Remote Base (here *compénsatory*).[1]

Examples of the dependency between stem allomorph inventory and whether markedness-reducing phonological processes apply to derived paradigm members have been noted in numerous languages (see section 1 of Breiss (2024) for an effort at a complete list at the time of writing). Although most commonly discussed in terms of a static correlation between stem allomorph shapes and the types of derived forms in the lexicon, a growing body of experimental work has demonstrated that speakers represent this correlation in their grammars, and extend it to novel forms (cf. Steriade & Stanton (2020) and Breiss (2024) on English, and Breiss (2024) on Spanish).

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[1] The distinction between types of Remote Base made in Breiss (2024) is not discussed here, as all Remote Bases have stems with phonological properties which, if adopted in the Derivative, would reduce markedness relative to the Local Base-faithful form, and so are in Breiss' terms "helpful".

2. An experiment to detect Base-specificity

Like the examples *cómpensate* and *ínundate* given in section 1, most previous experimental work has used Local Bases that only have two possible locations for stress, or more generally, only had a single option that could be pursued to resolve the markedness induced by affixation. For those studies, the question was whether speakers produced Derivatives for Local Bases that had a Remote Base that embodied that repair differently than for Local Bases that had no such licensing stem allomorph elsewhere in their paradigm. For example, the Local Base *cómpensate* has, when stripped of its affix *-ate*, only one alternative option for placing main stress on the stem, on what Breiss (2024) termed the "target syllable": *compéns-able*. Thus, it has gone untested whether the existence of a Remote Base simply decreases the odds of the Derivative having stress placement matching the Local Base, or whether it specifically has an *attractive* force that increases the odds of the Derivative having stress placement that *matches* that Remote Base. For example, consider the Local Bases *ánagram* (no Remote Base), *ánalog* (Remote Base *análogous* with stress on the second syllable of the stem) and *vítriol* (Remote Base *vitriólic* with stress on the third syllable of the stem), all creating novel Derivatives in *-ist*. The first scenario, where the presence of the Remote Base simply encourages deviation from the Local Base, predicts that both penult-stressed *análogist* and *vitríolist* should have equal probability and be greater in probability than penult-stressed *anágramist*, and that the finally-stressed *analógist* and *vitriólist* should be equally likely, and more likely than *anagrámist*. The second possibility, that Remote Bases exert a specific attractive pull on the Derivative to resemble them, predicts differing rates of probability for the derived forms with different types of Remote Bases, with *análogist* being more likely than *vitríolist*, and *vitriólist* being more likely than *analógist*. This a question that can only be addressed by looking at Local Bases that have more than one candidate target syllable.

2.1. Methods

54 participants were recruited from the UCLA SONA Psychology Subject Pool, and were compensated with course credit for their time. 15 were excluded for not having spoken English since before the age of seven consistently in some context (home, school, with certain family members, etc.), and 8 for technical problems relating to the sound quality, leaving 31 participants with data included in this study.

Local Bases included in this experiment were trisyllabic nouns with initial stress, and were selected to investigate whether the presence of a helpful Remote Base increases Derivative similarity to that Remote Base in particular, or simply decreases the odds that the Derivative is faithful to the Local Base. These 45 Local Bases were divided evenly into 15 Local Bases with no Remote Bases (ex., *ábacus, céntipede*), 15 Local Bases with a Remote Base with penult stress (ex., *ánalog - análogous, mónotone - monótonous*), and 15 Local Bases with Remote Bases with final stress (ex., *vítriol - vitriólic, sénator - senatórial*). The full list of stimuli is provided in the supplementary materials. A single affix, *-ist*, was selected based on the description in Marchand (1960) that it was neither stressed nor obligatorily stress-attracting (see also Aronoff 1976). Local Bases were recorded in isolation by a native speaker of North American English, and normalized to an intensity of 70 dB.

The experiment was conducted over the internet using the Labvanced experimental platform (Finger et al. 2017). At the start of the experiment, participants were told that they would be being asked to create new words of English, combining a word that they heard played through their speaker and an ending that was displayed on the screen, and speaking their response out loud. They were encouraged to seat themselves in a quiet location and make sure their speakers and microphone worked. They were also told that they would be asked to indicate whether they knew some of the words in the study, and that the criterion for "knowing" a word would be that they would be able to hear it used in a conversation and not need to stop to ask what it meant.

The experiment proceeded in the following way: after instructions and informed consent, a random half of the Remote Bases were presented visually, one at a time, in a vocabulary check phase. The participant indicated whether they knew the word via button press. Then the participant carried out the Derivative-formation trials: on each trial, they were instructed to click an onscreen button which would play the recording of the Local Base, while the affix *-ist* was shown on screen. The participant spoke their response — the Derivative, combining the Local Base and the ending — aloud, and clicked another button to proceed to the next trial. Participants were not allowed to proceed to the next trial unless they had played and completed the recording of the Local Base. After the production task, there was another vocabulary check which included the rest of the Remote Bases, and all the Local Bases. Finally, participants completed a short language background questionnaire.

Due to an error in the configuration of the randomization measures for the trials, each participant only saw a randomly-selected subset of ≈85% of the Derivative formation trials (median 30 trials per subject, standard deviation 5); vocabulary-check trials were not affected. Since the missing trials were distributed randomly among item types and subjects, I do not judge this to be a reason to believe the results of the experiment should be biased; however it is likely that parameter estimates in statistical models will have greater uncertainty than they otherwise might have.

2.2. Data annotation, exclusion, and analysis

Each Derivative production was coded for whether the participant knew the Local Base, and whether they knew the Remote Base if there was one. In this way, localized the effect of the Remote Base to the individual speaker, rather than assuming that each speaker knows all the relevant Remote Bases. Derivatives formed to Local Bases that were not known were excluded, and the primarily-stressed syllable for each remaining Derivative was recorded as the dependent variable; if participants produced more than one Derivative on a given trial, the last one was coded as the response. Trials where the Derivative was inaudible or primary stress could not be determined due to audio recording issues were excluded. This left 914 trials whose Derivatives were analyzed here.

The data was analyzed using a multinomial mixed effects Bayesian regression model, fit using the *brms* package (Bürkner 2017) in the R computing environment. Since the multinomial model is parameterized in terms of two binary logistic regressions, the model had two intercepts and two sets of fixed effects, one corresponding to the log-odds of a Derivative being penult-stressed vs. being initial-stressed (faithful to the Local Base), and the other corresponding to the log-odds of a Derivative being final-stressed vs. initial-stressed. The model contained a three-level fixed effect of Remote Base stress location for the speaker who produced the Derivative: none (ex., *ánagram*, coded as the intercept), penult-stressed (ex., *ánalogue - análogous*) and final-stressed (ex., *sénator - senatórial*). The model also contained a fixed effect of whether the Local Base had secondary stress on the final syllable or not (ex., *óctagòn* vs. *gélatin*). The model contained a random intercept for Local Base and a random intercept for subject, with random slopes of all fixed effects by subject. The model used default, weakly-informative priors on all parameters.

A Bayesian model returns samples from the posterior over probable values for parameters in light of the priors and the data, rather than a point estimate. Therefore we report the *median* of these values as the measure of central tendency, along with values that encompass the central 95% of the probability mass of the distribution (termed 95% Credible Intervals, or 95% CrI). We also report the probability of a directional effect, *p*-direction, which is the proportion of the posterior that falls to one side of zero in the direction of the parameter coefficient. This measure ranges from 0.5 (when the posterior is centered on zero, with as much evidence for a positive effect as a negative one) to 1 (when the posterior excludes zero entirely, indicating a strong effect in the direction of the parameter coefficient). For more details on Bayesian models in the linguistic context, see Nicenboim & Vasishth (2016), Nalborczyk et al. (2019), and Vasishth et al. (2018).

2.3. Results

To restate the empirical question, we are interested in whether Remote Bases encourage Derivatives to *match* their stress, or simply to be unfaithful to the Local Base without specifically matching the stress placement of their own Remote Base. We can examine this by plotting the probability of stress placement on each of the syllables of the stem of the Derivative divided by the type of Remote Base that exists for that participant, shown in figure 2.3.

Turning first to the Derivatives in the leftmost panel, which have no Remote Bases, we observe what we can think of as the “default” behavior of the phonological grammar in the face of the affix *-ist* being attached to a trisyllabic, stress-initial monomorpheme. Although the majority of the time the Derivative

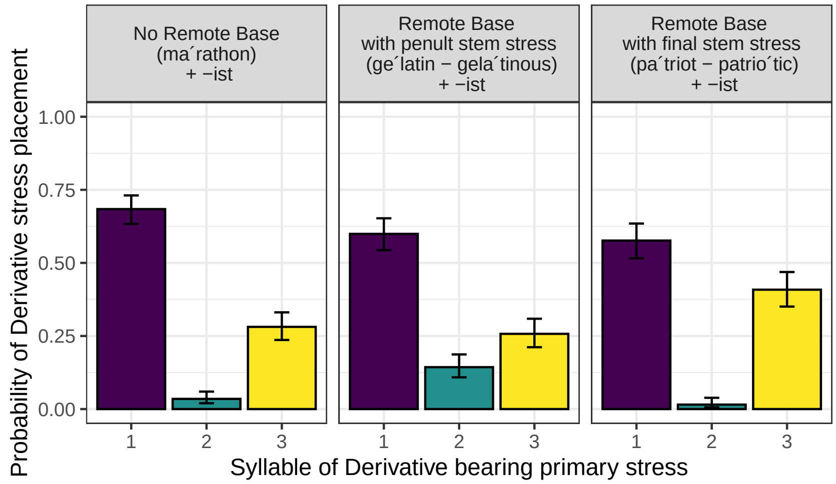

Figure 1: Results of the Derivative production task with trisyllabic Local Bases. The vertical axis plots the mean and binomial confidence interval of the probability of the Derivative having stress on a given syllable of the Derivative; the syllable of the Derivative is plotted within each panel the horizontal axis; panels divide responses by stressed syllable in the Remote Base, if known to subject.

is faithful to the Local Base and retains initial stress, a substantial minority of cases have stem-final stress. This outcome is the best case of phonotactically-driven repair, allowing the amelioration of the long lapse in initially-stressed Derivatives, while retaining secondary stress on the initial syllable, and leaving the vowel qualities of the first and second syllables unchanged. In only a very small handful of cases do participants place primary stress on the second syllable, an outcome which is harmonically-bounded by the fully-faithful initial stress placement and the fully-lapse-repairing final stress placement.[2]

Moving on to the center and right panels of figure 2.3, we find that the common, but previously untested, assumption about the role of the Remote Base in Derivative formation is borne out. Derivatives with Remote Bases, when not faithful to the Local Base, are more likely to resemble their particular Remote Base in terms of stress placement. That is, Derivatives to Local Bases with penult-stressed Remote Bases, like the Local-Remote pair *ánalog - análogy*, have a greater proportion of stress on the corresponding, penultimate syllable of the Derivative than those Derivatives formed to Local Bases without Remote Bases (i. e., more *análogist* than *maráthonist*) and to those with final-stressed Remote bases. This makes sense from the point of view of the penult-stressed Derivative being supported by the presence of a Remote Base that shares its stress location. Correspondingly, Derivatives to Local Bases with finally-stressed Remote Bases have more non-faithful stress placement on their final stem syllable than Derivatives to Local Bases with no Remote Base or ones with penult stress; that is, for the Local-Remote Base pair *vítriol - vitriólic*, more *vitriólist* than *marathónist* and *anagrámist*. These effects are statistically robust, as described in table 1.

Looking at the statistical model, we can see that knowledge of the Remote Base meaningfully drives Derivative stress placement: if the Derivative has a penult-stress Remote Base, then the probability of penult stress is higher than when there is no Remote Base. The same is true for Derivatives with final-stressed Remote Bases. Above and beyond the effect of the Remote Base, secondary stress on the final syllable of the Local Base (as in *ántelòpe*) reliably attracts stress in the Derivative.

[2] Though not the focus of this paper, I note in passing that the non-zero probability of penult stress in Derivatives to trisyllabic Local Bases with no Remote Bases constitutes an empirical case of *gradient harmonic bounding*, predicted by MaxEnt approaches but impossible under the standard Noisy / Stochastic Harmonic Grammar models (Hayes 2017: cf.).

Parameter	*Median*	*95% CrI*	*p*-direction
Intercept (initial stress vs. penult stress):			
Stress profile = *100*			
Know Remote Base = *no*	-5.56	[-7.78, -3.92]	
Stress profile = *102*	1.33	[0.17, 2.70]	0.99
Know Remote Base with penult stress = *yes*	1.63	[-0.07, 3.35]	0.97
Know Remote Base with final stress = *yes*	-2.15	[-7.96, 0.94]	0.87
Intercept (initial stress vs. final stress):			
Stress profile = *100*			
Know Remote Base = *no*	-2.69	[-3.67, -1.78]	
Stress profile = *102*	1.99	[1.20, 2.84]	≈1
Know Remote Base with penult stress = *yes*	-0.06	[-0.77, 0.64]	0.57
Know Remote Base with final stress = *yes*	0.67	[-0.05, 1.39]	0.97

Table 1: Results of the Bayesian model fit to Derivative productions. Coefficients are in log-odds, with positive signs indicating an increase in stress shift vs. the intercept. Stress profiles are given using the SPE stress numbering system (Chomsky et al. 1956, Chomsky & Halle 1968): *1* denotes primary stress, *2* a secondary stress, and *0* an unstressed syllable.

3. Modeling the experiment

We now turn to formal modeling of the experimental data. This is important not only to obtain a better understanding of the experimental data in light of phonological theory — as the phonological model allows us to test theoretical claims that do not have any direct correlate in the statistical model — but it also allows us to select between these theories on the basis of the evidentiary support they gain from the experimental data. Below, we fit three phonological models to the experiment, each implementing a distinct proposal for how speakers' grammars represent Lexical Conservatism. In an effort to render the comparison as clean as possible we focus specifically on the role of the Remote Base, abstracting for the moment away from the effect of secondary stress in the Local Base. In terms of grammar formalism, because of the non-categorical nature of the data and in light of the findings of Breiss (2024: section 2.2), we use Maximum Entropy (MaxEnt) grammars to implement all three analyses.

3.1. The Voting Bases model of Breiss (2024)

This model directly implements the "attractive" force demonstrated in the experiment above by using multiple faithfulness constraints, one for each Base: ID-LOCAL and ID-REMOTE. The model shares with the other two assessed below the assumption that all stem allomorphs — Bases — are accessible to the grammar as inputs. Specific to the Voting Bases model is the principle that each candidate is penalized by multiple faithfulness constraints, each to the extent that the candidate does not resemble the Base that the specific faithfulness constraint refers to. We also ise *LAPSE-RIGHT (abbreviated *LAPSE), common to all three models, which assigns a violation to each pair of adjacent syllables without primary stress at the right edge of a word, to enforce the markedness penalty that drives deviation from faithfulness to the Local Base. Table 3.1 shows this model fit to the experimental data; the candidates are examples chosen for illustrative purposes.

3.2. The quantification-based approach of Steriade (1997)

This model takes a different approach, where the presence of a Remote Base in the lexicon merely *licenses* the deviation of the Derivative from the Local Base. In the model of the experimental data in table 3.2, LEX-[stress] quantifies over all stress positions in the Local Base's paradigm, and assesses a violation only if there is no Remote Base that has a stress placement that matches that candidate's stress.

márathon (L) Weight:	ID-LOCAL 4.29	ID-REMOTE 0.84	*LAPSE 1.66	*p*
a. *márathonist*			2	.69
b. *maráthonist*	1		1	.05
c. *marathónist*	1			.26
gélatin (L), *gelátin-* (R)	ID-LOCAL	ID-REMOTE	*LAPSE	
a. *gélatinist*		1	2	.65
b. *gelátinist*	1		1	.11
c. *gelatínist*	1	1		.24
sénator (L), *senatór-* (R)	ID-LOCAL	ID-REMOTE	*LAPSE	
a. *sénatorist*		1	2	.51
b. *senátorist*	1	1	1	.04
c. *senatórist*	1			.45

Table 2: Tableaux with fitted weights for the Voting Bases model; inputs are examples of their paradigm type, parenthetical L and R denote Local and Remote bases in the input.

márathon (L) Weight:	LEX-[stress] 1.29	*LAPSE 0	*p*
a. *márathonist*		2	.64
b. *maráthonist*	1	1	.18
c. *marathónist*	1		.18
gélatin (L), *gelátin-* (R)	LEX-[stress]	*LAPSE	
a. *gélatinist*		2	.44
b. *gelátinist*		1	.44
c. *gelatínist*	1		.12
sénator (L), *senatór-* (R)	LEX-[stress]	*LAPSE	
a. *sénatorist*		2	.44
b. *senátorist*	1	1	.12
c. *senatórist*			.44

Table 3: Tableaux with fitted weights for the quantification-based model; inputs are examples of their paradigm type, parenthetical L and R denote Local and Remote bases in the input.

3.3. The cyclicity-based approach of Steriade & Stanton (2020)

The final theory we evaluate uses a modified concept of the phonological cycle (Chomsky & Halle 1968) to capture the difference in behavior between Derivative types. Here, each candidate Derivative corresponds to a single Base in the input — Local *or* Remote — and a single faithfulness constraint penalizes unfaithfulness to that Base. To give faithfulness to the Local Base priority while still permitting the Remote Base to exert an influence to relieve phonological markedness, the model incorporates a constraint called CYCLIC CONTAINMENT (abbreviated C-CONTAINMENT), which penalizes a candidate Derivative for standing in correspondence to a non-Local Base. We use the same markedness constraint as the previous two analyses. Note that because some candidates differ only in the Base to which they correspond, we sum over these phonetically-indistinguishable possibilities when fitting the model.

3.4. Results of model fitting

Weights that maximized the likelihood of the experimental data were fitted using the *Solver* utility in Microsoft Excel (Fylstra et al. 1998), and the predicted probabilities of each of the types of Derivatives from the experiment are shown below in figure 2.

márathon (L)		BD ID-[stress]	C-CONT.	*LAPSE		
	Weight:	3.8	17.50	1.55	*p*	
a.	*márathonist* (L)			2	.62	
b.	*maráthonist* (L)	1		1	.07	
c.	*marathónist* (L)	1			.30	
gélatin (L), *gelátin-* (R)		BD ID-[stress]	C-CONT.	*LAPSE	*p*	sum *p*
a.	*gélatinist* (L)			2	.62	.62
b.	*gélatinist* (R)	1	1	2	≈0	
c.	*gelátinist* (L)	1		1	.07	.07
d.	*gelátinist* (R)		1	1	≈0	
e.	*gelatínist* (L)	1			.30	.30
f.	*gelatínist* (R)	1	1		≈0	
sénator (L), *senatór-* (R)		BD ID-[stress]	C-CONT.	*LAPSE	*p*	sum *p*
a.	*sénatorist* (L)			2	.62	.62
b.	*sénatorist* (R)	1	1	2	≈0	
c.	*senátorist* (L)	1		1	.07	.07
d.	*senátorist* (R)	1	1	1	≈0	
e.	*senatórist* (L)	1			.30	.30
f.	*senatórist* (R)		1		≈0	

Table 4: Tableaux with fitted weights for the cyclicity-based model; inputs are examples of their paradigm type, parenthetical L and R denote Local and Remote bases in the input, and in candidates the Base which the candidate stands in correspondence to.

A visual inspection of figure 2 reveals that the three theories differ substantially in their ability to model the experimental data. The Voting Bases theory (left) correctly captures the finding that for Derivatives with penult-stressed Remote Bases, more penult-stressed Derivatives are formed than for Local Bases with no Remote Bases and those with final-stress Remote Bases, and similarly for Derivatives with final-stressed Remote Bases compared to the proportion of final-stressed Derivatives for Local Bases with no Remote Bases and penult-stressed Remote Bases. That is, the model can capture both the consistency in overall *ranking* of the Derivative stem syllables within Remote Base type that receive stress (initial, final, penultimate), while at the same time capturing the fact that the actual proportion of times that each syllable receives stress differs corresponding to the Remote Base stress placement for that Derivative.

The other models fare less well: the quantification-based theory (center) distinguishes between the three types of Derivatives, but does so simply by penalizing those that do not resemble some Remote Base, this failing to capture the quantitative facts of the actual data pattern. Finally, the cyclicity-based account (right) falls prey to the opposite issue: it can quantitatively distinguish between the three stem syllables of the Derivative, but cannot capture the influences of the Remote Base across the three categories of Derivative. Thus, on a visual inspection, it appears that only the Voting Bases theory is sufficiently flexible to capture the data patterns observed in the experiment.

This impressionistic conclusion is quantitatively supported: using the sample-size-corrected Akaike Information Criterion (Burnham & Anderson 2002), AICc, we assessed the degree of mis-fit induced by modeling the data under the different theories. Differences of AICc between two models (ΔAICc) larger than 10 are considered to constitute strong evidence that the model with the algebraically lower AICc value is better supported by the data (Burnham & Anderson 2002, 2004). We find that the ΔAICc favors the Voting Bases model over the quantification-based model (ΔAICc = 246.15) and the cyclicity-based model (ΔAICc = 42.25).

4. Conclusion

This paper tested an assumption in the literature about Lexical Conservatism — that Remote Bases specifically yielded Derivatives that resembled them — and found that it was supported in a *wug*-test with

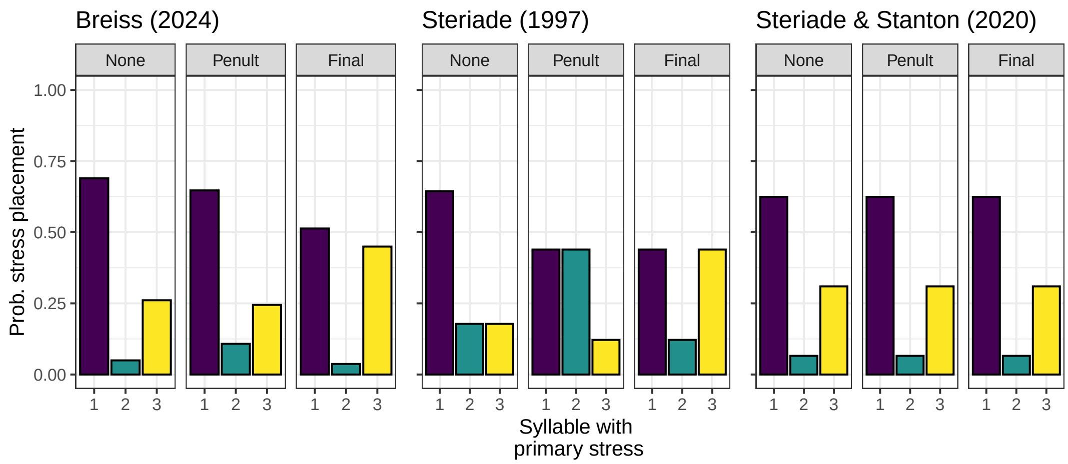

Figure 2: Plot of model-predicted probabilities for the three theories discussed above.

trisyllabic Local Bases. We then used this data as the empirical basis to compare three current models of Lexical Conservatism in the theoretical literature, and found that only the Voting Bases model of Breiss 2024 was flexible enough to capture the salient data patterns.

References

Aronoff, Mark. 1976. Word formation in generative grammar. *Linguistic Inquiry Monographs Cambridge, Mass* (1). 1–134.

Breiss, Canaan. 2024. When bases compete: a voting model of lexical conservatism. *Language* 100(2). 308–358.

Bürkner, Paul-Christian. 2017. Brms: an R package for Bayesian multilevel models using Stan. *Journal of statistical software* 80(1). 1–28.

Burnham, Kenneth P & David R Anderson. 2002. *Model selection and multimodel inference: a practical information-theoretic approach*. 2nd edn. Springer New York.

Burnham, Kenneth P & David R Anderson. 2004. Multimodel inference: understanding aic and bic in model selection. *Sociological methods & research* 33(2). 261–304.

Chomsky, Noam & Morris Halle. 1968. *The sound pattern of English*. Harper & Row New York.

Chomsky, Noam, Morris Halle & Fred Lukoff. 1956. On accent and juncture in english. *For Roman Jakobson* 65. 80.

Finger, Holger, Caspar Goeke, Dorena Diekamp, Kai Standvoß & Peter König. 2017. Labvanced: a unified javascript framework for online studies. In *International conference on computational social science (cologne)*.

Fylstra, Daniel, Leon Lasdon, John Watson & Allan Waren. 1998. Design and use of the microsoft excel solver. *Interfaces* 28(5). 29–55.

Hayes, Bruce. 2017. Varieties of noisy harmonic grammar. In *Proceedings of the annual meetings on phonology*, vol. 4.

Marchand, Hans. 1960. *The categories and types of present-day word formation*. Harassowitz, Weissbaden.

Nalborczyk, Ladislas, Cédric Batailler, Hélène Lœvenbruck, Anne Vilain & Paul-Christian Bürkner. 2019. An introduction to Bayesian multilevel models using brms: a case study of gender effects on vowel variability in standard Indonesian. *Journal of Speech, Language, and Hearing Research* 62(5). 1225–1242.

Nicenboim, Bruno & Shravan Vasishth. 2016. Statistical methods for linguistic research: Foundational Ideas — Part II. *Language and Linguistics Compass* 10(11). 591–613.

Steriade, Donca. 1997. Lexical conservatism. *Linguistics in the morning calm*. 157–179.

Steriade, Donca & Juliet Stanton. 2020. Productive pseudo-cyclicity and its significance. Talk at LabPhon 17. `https://julietstanton.github.io/files/steriade%5C_stanton%5C_labphon17.pdf`.

Vasishth, Shravan, Bruno Nicenboim, Mary E Beckman, Fangfang Li & Eun Jong Kong. 2018. Bayesian data analysis in the phonetic sciences: a tutorial introduction. *Journal of phonetics* 71. 147–161.